

CAREERRADAR

Problem Statement — Understanding Your Real Position in the Job Market

1. The Problem

For students, fresh graduates, and early-career developers, finding the right job is no longer just about searching for vacancies. The bigger challenge is understanding what the market actually expects from them right now.

Job opportunities are spread across different platforms and websites. A candidate may find dozens of roles with similar titles, but each role can ask for a different combination of technologies, experience, tools, and skills. This makes it difficult to answer a simple question: **“If I apply for these jobs today, how prepared am I?”**

Most existing job-search experiences focus on helping users discover jobs. They do not necessarily help a candidate understand the gap between their current skills and the skills repeatedly requested by the live job market.

2. Why This Matters

A student might know React, Node.js, and MongoDB and believe they are ready for a full-stack role. After looking at current openings, however, they may discover that many relevant positions also mention TypeScript, Docker, AWS, testing, system design, or other requirements.

The problem is not simply a lack of information. There is too much information, and it is difficult to turn that information into a clear personal action plan. Candidates need a way to connect **live job-market demand** with **their own skills**.

3. Core Problem Statement

How might we help students and early-career developers understand their current readiness for real job opportunities by analyzing live job-market requirements and translating those requirements into clear, actionable skill gaps?

4. What the Solution Needs to Address

- **Live job-market data:** The system should be able to discover current and relevant job opportunities rather than relying only on a static database.
- **Relevant opportunity discovery:** Users should be able to search based on a target role, location, or similar criteria.
- **Requirement understanding:** The system should identify the skills, technologies, and requirements appearing in relevant job listings.
- **Personal skill comparison:** A candidate should be able to provide their current skills and compare them against the requirements found in the market.
- **Clear readiness feedback:** The output should be understandable instead of presenting users with a long list of raw job descriptions.
- **Market-level insight:** The system should identify skills that appear repeatedly across multiple relevant opportunities, not just one job.

- **Actionable next steps:** The user should be able to understand which skills or areas deserve attention next.

5. Intended Users

The primary users are college students, fresh graduates, self-taught developers, and early-career professionals who are preparing to enter or move within the technology job market. The concept can also be useful for learners who want evidence-based direction about which skills to prioritize next.

6. Expected User Experience

The experience should feel less like another job board and more like a personal market-readiness check. A user should be able to provide a target role and their current skill set, then receive information derived from current job opportunities.

User provides	System should understand
Target role	What type of opportunities to analyze
Location / preference	Where relevant opportunities are available
Current skills	What the candidate already knows
Live job results	What the market is currently asking for

7. Key Questions the Product Should Answer

- What relevant jobs are available right now?
- What skills are those jobs repeatedly asking for?
- Which of those skills does the candidate already have?
- Which important skills are missing or weak?
- Which skills appear most frequently across the current market?
- What should the candidate focus on next based on the evidence?

8. Role of Live Search Data

Live search data is central to the problem because the technology job market changes continuously. A useful system should therefore be able to discover current opportunities and analyze the requirements attached to them. For this hackathon, SerpApi can provide the live search layer needed to access relevant job-search data.

The important requirement is that search data should not be included merely as a technical checkbox. The product should depend on fresh search results to produce its insights. If the live market changes, the product's analysis should be able to change with it.

9. What Success Looks Like

A successful solution should leave a user with more than a list of job links. After using it, the user should have a clearer picture of the opportunities available to them, the skills currently valued in those opportunities, and the areas they may want to improve.

The goal is not to tell a candidate which job they should take. The goal is to give them better information about the market so they can make a more informed career decision.

10. Constraints & Scope for the Hackathon

- The solution should be practical enough to demonstrate as a working prototype.
- The core workflow should be understandable within a short product demonstration.
- Live search data should be meaningfully integrated into the core experience.

- The prototype should prioritize one complete user journey over a large number of disconnected features.
- AI, if used, should support analysis or explanation rather than replace the entire product logic.

The Solution Is Intentionally Not Defined Here

This document defines the problem, users, requirements, and expected outcomes — not the implementation. The product architecture, user interface, search workflow, matching approach, AI components, and final experience are left open for exploration during development.

Build question for the live session:

Can we turn the live job market into a practical signal that helps a developer understand where they stand — and what to do next?